However microorganisms can be harmful in many ways and bring about undesirable changes. These microorganisms can cause diseases that can make us sick or even kill us. Although much more is known today about microbial life than ever before, the vast majority of this invisible world remains unexplored. Microbiologists continue to identify new ways that microbes benefit and threaten humans.

Microbiology is the study of living organisms of microscopic size, which include bacteria, fungi, algae, protozoa, and viruses. Microbiology is concerned with form, structure, reproduction, physiology, metabolism, and classification of microorganisms. It includes the study of

- their distribution in nature,
- their relationship to each other and to other living organisms,
- their effects on human beings, animals and plants,
- their abilities to make physical and chemical changes in our environment,
- their reaction to physical and chemical agents.

1.1 Groups of Microorganisms

There are many kinds of microorganisms present in the universe. They are broadly classified into the following groups.

Bacteria: They are unicellular prokaryotic organisms or simple association of similar cells. Cell multiplication usually happen by binary fission.

Example: *Escherichia coli, Bacillus subtilis*

Fungi: They are eukaryotic organisms which is devoid of chlorophyll. They are usually multicellular. They range in size and shape from single celled microscopic yeasts to giant multicellular mushrooms and puffballs.

Example: *Aspergillus niger, Agaricus bisporus*

Protozoa: They are unicellular eukaryotic organisms. Their role in nature are varied. The best known protozoa cause disease in human beings and animals.

Example: *Giardia lamblia, Plasmodium vivax*

Algae: They range from unicellular, colonial to multicellular forms. All algal cell contain chlorophyll and are capable of photosynthesis. They are found most commonly in aquatic environments and damp soil.

Example: *Spirogyra, Chlamydomonas*

Viruses: In the study of Microbiology, we encounter "organisms" which may represent the borderline of life. Viruses are simpler in structure and composition than other living cells. A virus is made up of nucleic acids and proteins. Viruses are obligate parasites. They grow only within an appropriate host cell (plant, animal, humans or microbe). They cannot multiply outside a host cell.

Example: HIV, Rabies virus



Prions are infectious agents composed entirely of protein material. Creutzfeldt–Jacob Disease (CJD) is one of the human prion diseases.

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